

Can We Trust OpenStreetMap to Measure Cycling-Infrastructure Change?

A Reproducible Validation Framework Using Google Street View

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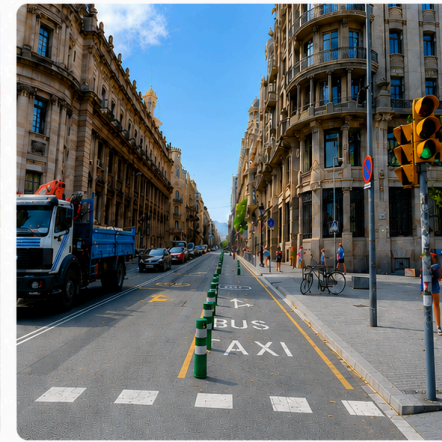
GEMOTT, Universitat Autònoma de Barcelona

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Why does this matter?

- Cities are rapidly expanding cycling infrastructure
- We need *longitudinal* data to understand its impacts
- OpenStreetMap (OSM) offers global, historical data
- But can we trust OSM to detect *change* over time?



Research aim

**Quantify the accuracy of OSM for detecting cycling-
infrastructure change**

Study setting and data

Study setting

- Barcelona, Spain
- 2015–2023

Data sources

- Historical OSM cycling-infrastructure snapshots
- Census-tract boundaries
- Historical Google Street View (GSV) imagery



Analytical workflow

1. Detect OSM cycling-infrastructure change



2. Stratify census-tract sample



3. Generate network validation points



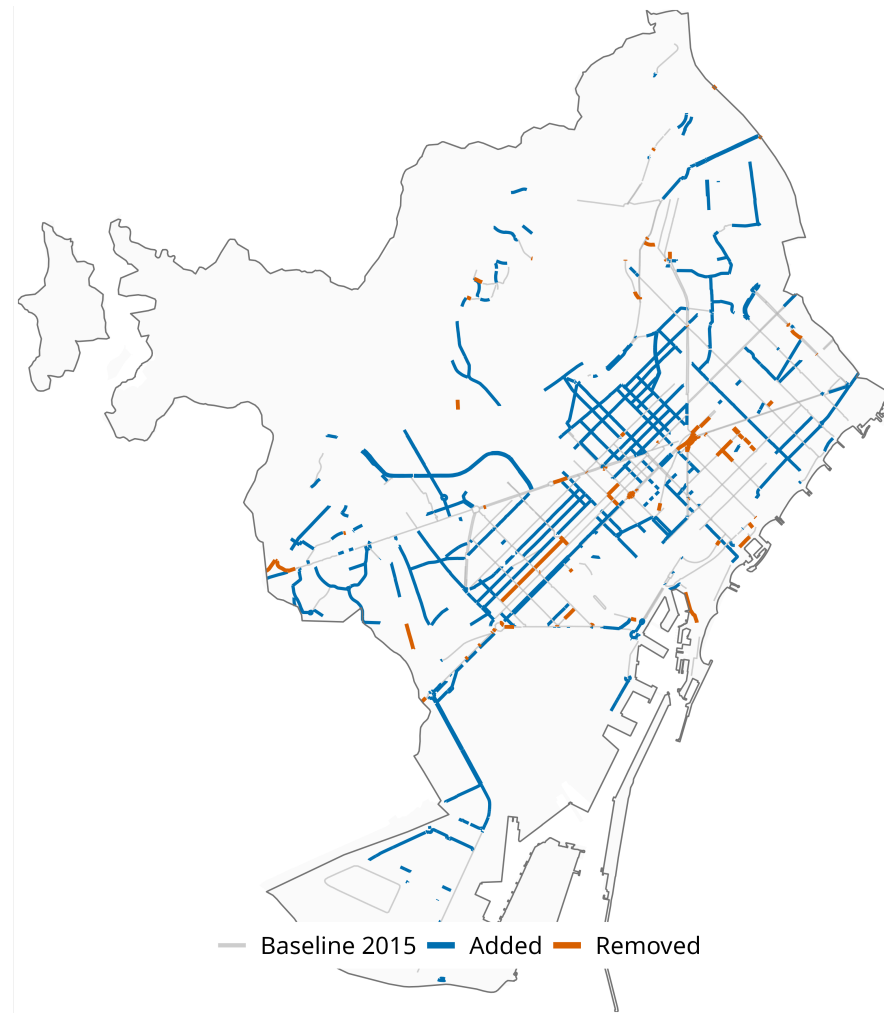
4. Inspect and code historical GSV imagery



5. Calculate validation metrics

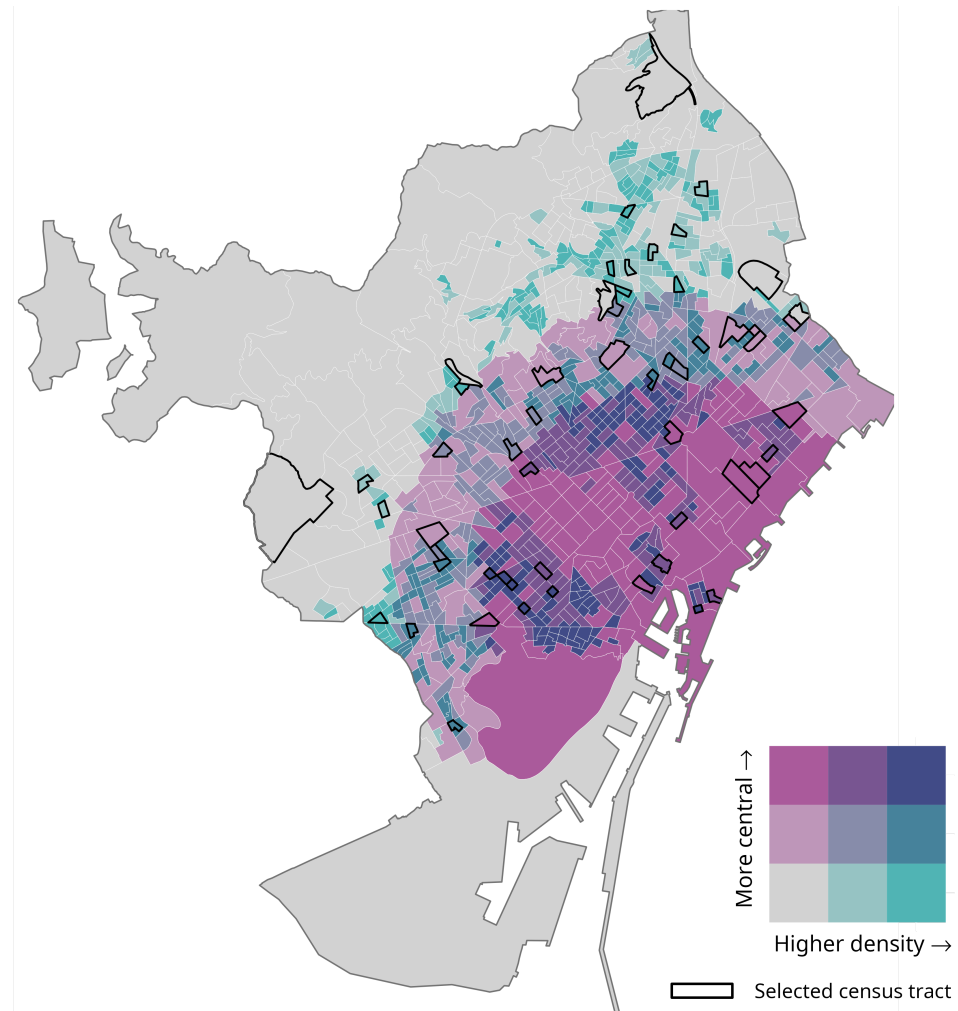
1. Detect OSM cycling-infrastructure changes

- OSM differencing identified:
 - 127 km additions
 - 25 km removals
- Net network growth:
 - +80%
- 946 candidate change segments



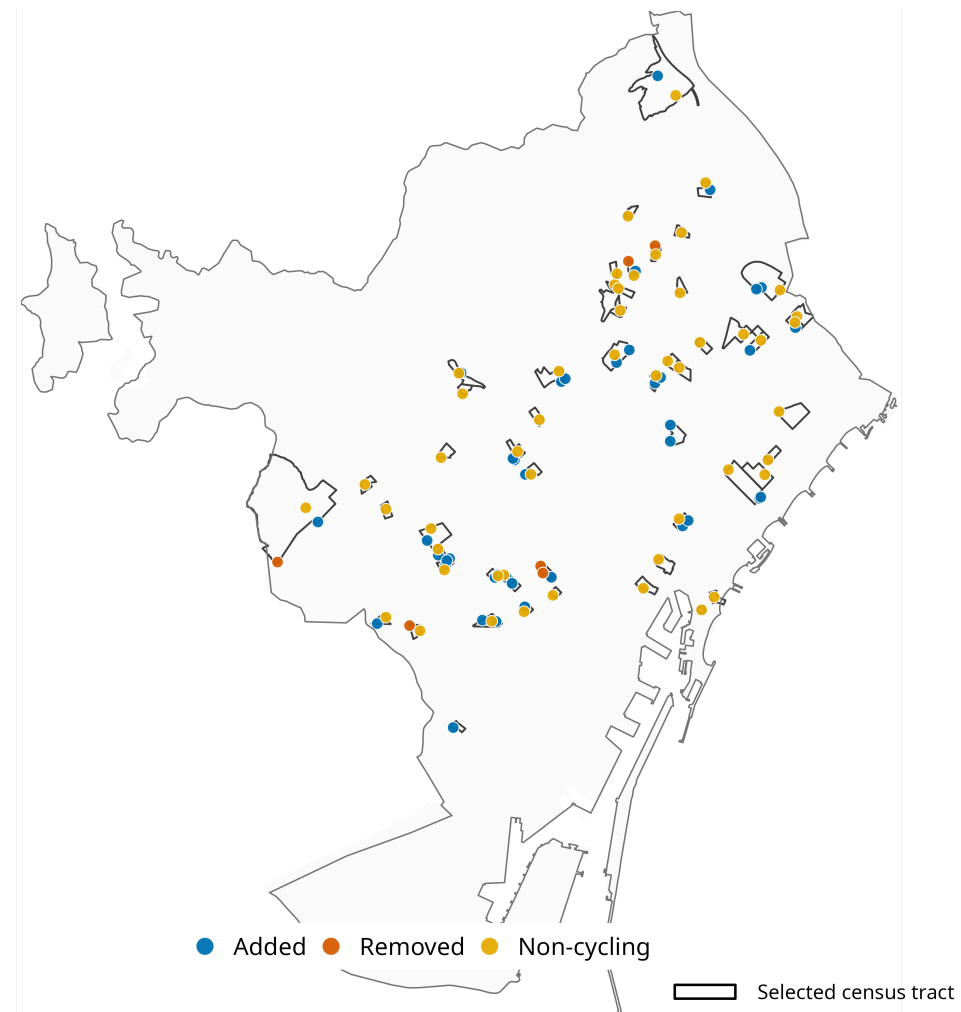
2. Stratify census-tract sample

- Barcelona stratified by:
 - population density
 - centrality
- 3×3 density–centrality strata
- 6 census tracts randomly sampled per stratum (54 total)



3. Generate network validation points

- Validation points sampled within selected census tracts
- Up to:
 - 2 additions
 - 2 removals
 - 1 non-cycling control
- Points linked to historical GSV imagery



4. Inspect and code historical GSV imagery



BEFORE (2015)



AFTER (2023)

Addition (0→1): newly implemented segregated cycle track



BEFORE (2015)



AFTER (2023)

Removal (1→0): cycle track removed after street redesign

5. Calculate validation metrics

$$\text{Precision} = \frac{TP}{TP+FP}$$

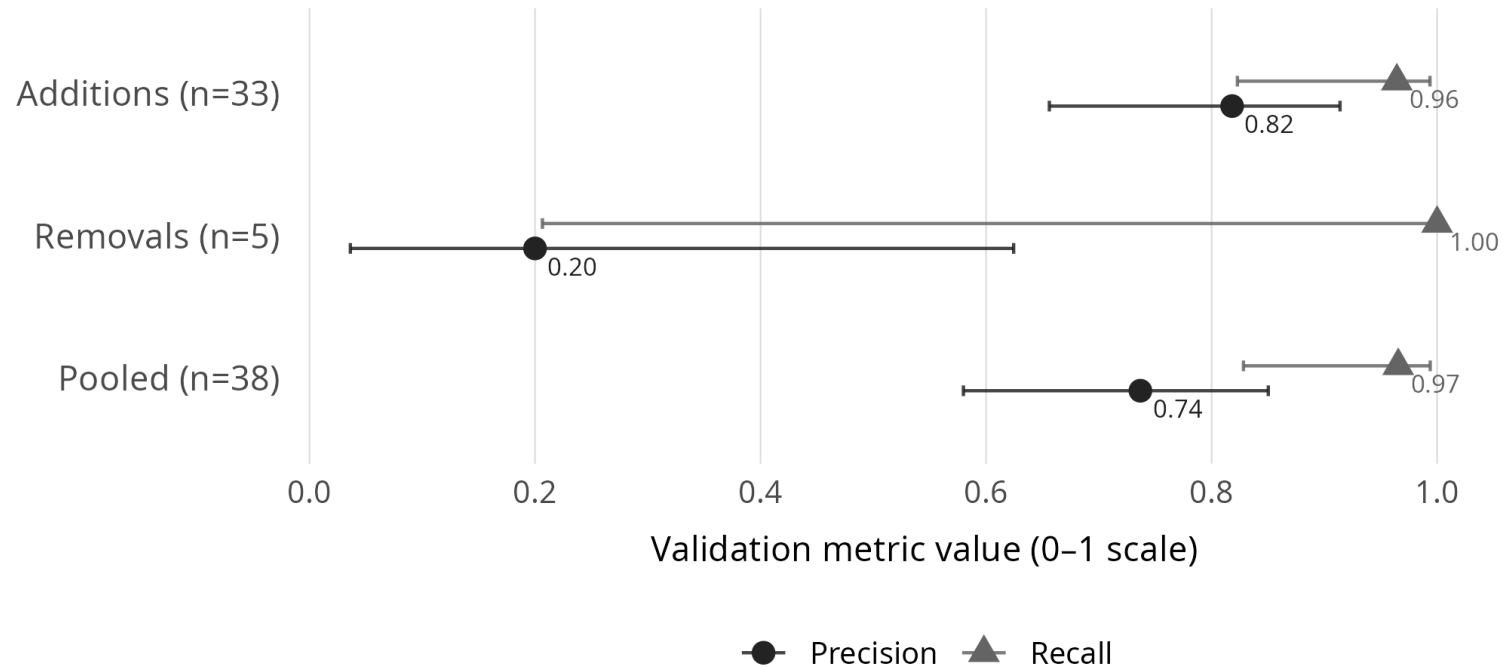
What proportion of detected OSM changes are real?

$$\text{Recall} = \frac{TP}{TP+FN}$$

What proportion of real changes are captured by OSM?

RESULTS

Validation performance



- High precision and recall for additions
- Most apparent removals were not confirmed; estimates remain uncertain (small sample)

Limitations

- GSV availability varies spatially and temporally
- Manual validation remains necessary
- Removal estimates based on small sample sizes
- Performance may vary across cities and OSM mapping communities

Conclusions

- Additions are detected reliably
 - Useful for tracking cycling-network expansion
- Removal estimates remain uncertain
 - Often reflect mapping edits rather than real infrastructure loss
- Findings may not generalise automatically to other cities
 - Barcelona is likely a relatively favourable case
 - Validation remains important in other contexts
- Reproducible framework to measure, evaluate, and calibrate OSM-based change

Thank you · 谢谢

Questions?

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Code and reproducible workflow available upon request